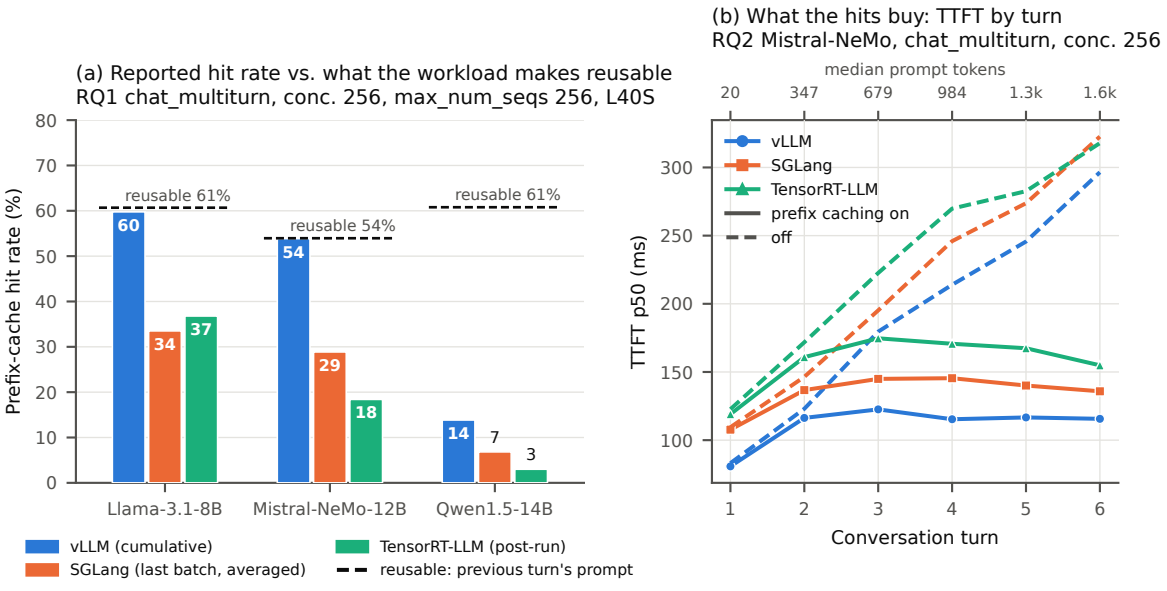
(a) Reported hit rate vs. what the workload makes reusable
RQ1 chat_multiturn, conc. 256, max_num_seqs 256, L40S
Prefix-cache hit rate (%)
80
70
60
50
40
30
20
10
0
reusable 61%
reusable 54%
reusable 61%
60
34
37
54
29
18
14
7
3
Llama-3.1-8B
Mistral-NeMo-12B
Qwen1.5-14B
vLLM (cumulative)
SGLang (last batch, averaged)
TensorRT-LLM (post-run)
reusable: previous turn's prompt
(b) What the hits buy: TTFT by turn
RQ2 Mistral-NeMo, chat_multiturn, conc. 256
median prompt tokens
20
347
679
984
1.3k
1.6k
TTFT p50 (ms)
300
250
200
150
100
vLLM
SGLang
TensorRT-LLM
prefix caching on
off
1
2
3
4
5
6
Conversation turn